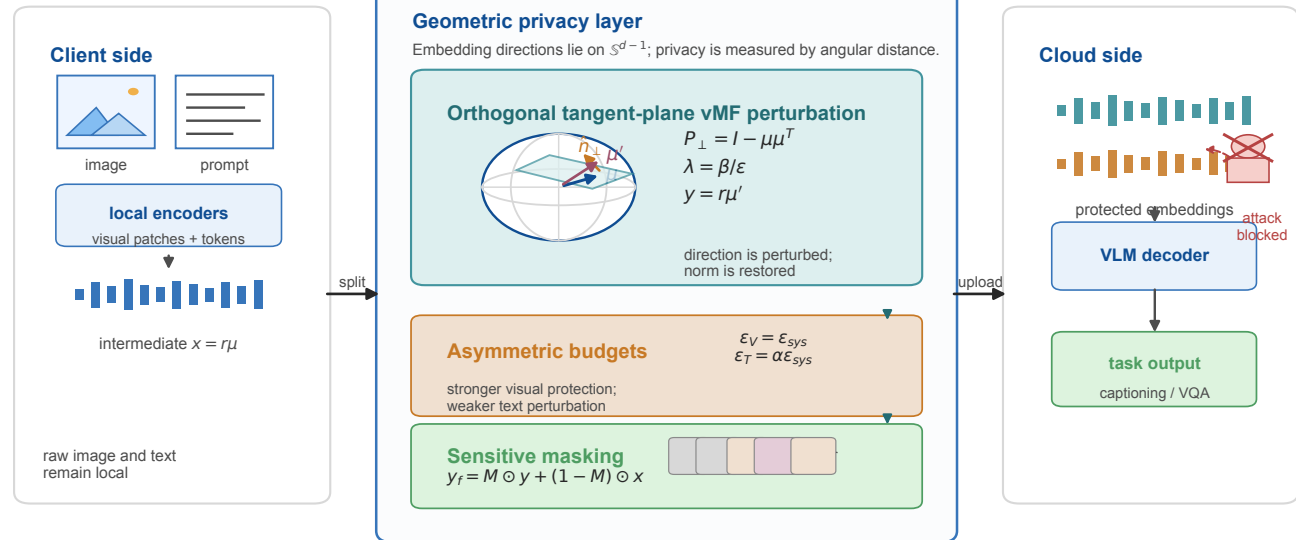


Semantic-preserving metric privacy for split VLM inference

Client-side vMF perturbation protects intermediate embeddings while preserving task semantics.

a End-to-end privacy layer in split VLM inference



b Norm-preserving geometry

Perturbation changes direction, not activation scale.
 $\|y\|_2 = \|x\|_2$ stabilizes LayerNorm and attention.



c Approximate metric privacy

Nearby semantic directions have close output laws.
 $\Pr[M(x) \in S] \leq e^{Kd_{\text{geo}}} \Pr[M(x') \in S] + 2\delta_{TV}$.



d Utility-aware protection

Asymmetric budgets and token masks protect private user content while preserving system prompts and downstream utility.

